# CMSE 801 - Homework 1 Rubric

In this homework assignment, students are primarily expected to practice their python fundamentals. In particular, students will be working with lists, dictionaries, loops, if statements, and functions. Note that they will have only very recently been exposed to functions.

The total number of points possible in this assignment is: **22**

## Question 1: Working with lists and dictionaries (5 points)

### Part 1: Lists (4 points)

- The student should receive:
  - **1 point** for successfully writing code for bullets 2 through 5 (4 points total).

### Part 2: Dictionaries (1 point)

- The student should receive:
  - **1 point** for correctly creating the dictionary.

## Question 2: Find the nucleon mass (3 points)

- The student should receive:
  - **2 points** for successfully iterating through the list and computing the effective mass (while also printing the results).
  - **1 point** for computing the average for the last 5 masses.

## Question 3: Fibonacci Numbers (3 points)

- The student should receive:
  - **1 point** for having a working loop.
  - **1 point** for storing the sequence as a list.
  - **1 point** for correctly computing a new list of the ratios.

## Question 4: Carbon Dating (3 points)

- The student should receive:
  - **1 point** for correctly initializing the variables.
  - **1 point** for writing a functioning while loop.
  - **1 point** for finding and printing the correct time for to reach 35% of original amount.

## Question 5: Finding divisors (3 points)

- The student should receive:
  - **1 point** for iterating over the correct range of numbers
  - **1 point** for correctly using the modulo operator along with the logical "and".
  - **1 point** printing the right number of numbers.

## Question 6: Resolving disagreemnt between grad students (8 points)

- The student should receive:
  - **2 points** for writing a properly defined function wthat computes the ABV.
  - **2 points** for defining a function that takes in the "beers" dictionary and returns a list.
  - **2 points** for implementing the correctly logic for checking the accuracy of ABV calculations.
  - **2 point** for calling the second functions and printing the correct results.